

■ Python bytearray Tutorial — Mutable Sequence of Bytes

A bytearray in Python is a mutable sequence of integers in the range 0–255. It's similar to bytes, but allows element modification after creation.

■ Example Code:

```
# Creating a bytearray
# Index Positions:    0    1    2    3    4
data_list = bytearray([1, 2, 3, 4, 5])
# Negative Indexing: -5  -4  -3  -2  -1
print("Original bytearray:", data_list)

# ■ Item assignment (mutable)
data_list[0] = 100
print("After modifying first element:", data_list)

# ■ Reading elements by index
print("Element at index 0:", data_list[0])
print("Element at index 1:", data_list[1])
print("Element at index 2:", data_list[2])

# ■ Accessing using valid and invalid indexes
print("Element at index 3:", data_list[3])    # Valid
print("Element at index -1:", data_list[-1])  # Valid

# ■ Accessing out of range raises an error
try:
    print(data_list[-6])    # Invalid index
except IndexError as e:
    print("Error:", e)
```

■ Key Points

Feature	bytearray	Description
Type	Mutable sequence	Can be changed after creation
Value range	0–255	Only byte values allowed
Indexing	■ Yes	Both +ve and –ve supported
Iteration	■ Yes	Works like list iteration
Mutability	■ Mutable	You can assign new values